Pythia training dynamics: Adam adaptive directions

Pythia-14M Pythia-31M Pythia-70M

Updates 1-100 | loss N=5

Updates 100-400 | loss N=21

Change in ln(training loss)

Projected direction change $q_t^{(d)}$

Adam-direction cosine $\rho_t^{(d)}$

Adam direction $\|d_t\|_2$

Optimizer update

Optimizer update

$d_t = \widehat{m}_t/(\sqrt{\widehat{v}_t} + \varepsilon)$; $q_t^{(d)} = 1 - \rho_t^{(d)} \|d_t\|/\|d_{t-1}\|$ (direction change, not measured curvature).

Loss change only: N=5 left, N=21 right; $\ln L_t - \ln L_{t-1}$. Other rows unsmoothed. Seed 1235. LR and decay excluded from d.

Y limits vary by column; norm row uses log Y. Update 100 appears twice; loss smoothing spans the boundary.

Norms start at update 1; direction pairs at 2; left loss means at 6. No extra averaging of Adam directions.